

The DESI DR2 CMB+BAO Evidence for Dynamical Dark Energy is a Projection Artifact of the CMB Geometric Degeneracy

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ABSTRACT

The Dark Energy Spectroscopic Instrument (DESI) DR2 data, combined with Planck CMB and Type Ia supernovae, yield a reported 3.1σ preference for dynamical dark energy (w_0w_a CDM) over Λ CDM. Focusing on the CMB+BAO subset where the tension is approximately 2.3σ , we find this signal is a projection artifact of the CMB geometric degeneracy. Constraint residual eigendecomposition—isolating independent directions of covariance tightening between datasets—reveals the DESI-Planck tension is strictly one-dimensional: PC1 > 99.99% of the total covariance contraction, aligned with the $H_0 \leftrightarrow \omega_{\text{cdm}}$ geometric degeneracy direction $(+0.744\sigma, -0.576\sigma, -0.304\sigma)$ in $(H_0, \omega_{\text{cdm}}, \omega_b)$. A chi-squared scan along this direction reveals χ^2_{DESI} drops from 32.68 (13 dof, $p = 0.0019$) at the Planck posterior mean to 21.25 ($p = 0.068$) at $+1\sigma$, and 13.88 ($p = 0.382$) at $+2\sigma$, representing a $\Delta\chi^2 = 22.16$ improvement from a shift entirely within Λ CDM. The geometric degeneracy shift alone yields a larger χ^2 reduction than the w_0w_a extension ($\Delta\chi^2 = 9.7$, for 2 extra parameters, Wilks-corrected to $\sim 2.6\sigma$). Three lines of evidence support the artifact interpretation: the w_0w_a parameters are 66.9% degenerate with the geometric direction; fitting a 1D signal with a 2-parameter extension inflates the apparent significance; and 57% of randomly constructed 2-parameter extensions with geometric projection produce $> 3\sigma$ evidence in Monte Carlo simulation. All code and data are publicly available.

Keywords: Cosmological parameters — dark energy — large-scale structure of universe — cosmic microwave background — methods: statistical

1. INTRODUCTION

The Dark Energy Spectroscopic Instrument (DESI) recently reported a 3.1σ preference for dynamical dark energy over a cosmological constant, based on the full combination of DESI DR2 BAO, Planck CMB, and Type Ia supernovae (DESI Collaboration 2025). When restricted to the CMB+BAO subset—the focus of the present analysis—the tension is approximately 2.3σ (Leizerovich et al. 2025). The w_0w_a CDM (CPL) parameterization (Chevallier & Polarski 2001; Linder 2003) introduces two additional free parameters—the present-day equation of state w_0 and its time

derivative w_a —relative to the standard six-parameter Λ CDM model. The DESI best-fit values $w_0 \approx -0.45 \pm 0.18$, $w_a \approx -1.34 \pm 0.39$ have attracted significant attention as potential evidence for evolving dark energy.

Several independent analyses have questioned aspects of this claim. [Efstathiou \(2025\)](#) rotated BAO distances into perpendicular and parallel components, finding $w(z = 0.5) = -0.996 \pm 0.046$ consistent with a cosmological constant, and concluding DESI’s statistical approach gives a “misleading impression” of evolving dark energy. [Leizerovich et al. \(2025\)](#) developed a generalized tension metric based on eigendecomposition of the dispersion tensor, finding the effective tension drops from the nominal 2.3σ to 1.45σ after accounting for dimensionality. Other groups have noted internal tensions within the BAO+CMB+SN combination ([Wang & Mota 2025](#)), absence of dynamical dark energy signals in full-shape modeling ([O Colgain et al. 2025](#)), and parameterization bias in the CPL form ([Dinda et al. 2025](#); [Lee 2025](#)).

However, none of these critiques identify the simplest statistical issue: the DESI-Planck tension lives along a single direction—the well-known CMB geometric degeneracy. This matters because fitting a 1D signal with a 2-parameter extension inevitably inflates the apparent significance, via the look-elsewhere effect and parameter-space projection. The 3.1σ $w_0 w_a$ result, we argue, is a predictable consequence of this mismatch between the tension’s true dimensionality and the model’s degrees of freedom.

2. METHOD

2.1. The Constraint Residual Framework

We employ constraint residual decomposition, a formalism for extracting independent directions of covariance contraction when a second dataset is added to a first ([Sam & Deng 2026](#)). Given two datasets $\mathcal{D}_1$ (Planck CMB) and $\mathcal{D}_2$ (DESI BAO), we consider the parameter covariance matrices $\Sigma_P = \text{Cov}[\theta|\mathcal{D}_1]$ and $\Sigma_{PD} = \text{Cov}[\theta|\mathcal{D}_1, \mathcal{D}_2]$. The constraint residual operator:

$$\mathcal{B} = \Sigma_P^{-1/2} (\Sigma_P - \Sigma_{PD}) \Sigma_P^{-1/2} \quad (1)$$

captures the additional constraining power of $\mathcal{D}_2$ beyond $\mathcal{D}_1$. Its eigendecomposition $\mathcal{B} = \sum_i \lambda_i \mathbf{v}_i \mathbf{v}_i^T$ yields principal constraint directions $\mathbf{v}_i$ with eigenvalues λ_i representing the variance reduction along each direction. The number of significant eigenvalues ($\lambda_i \gg 0$) determines the true dimensionality of the tension.

2.2. Data and Sampling

Planck CMB: We use the Planck 2018 baseline 6-parameter Λ CDM Markov Chain Monte Carlo (MCMC) chain (plikHM_TTTEEE_lowl_lowE_lensing) with $N = 20,000$ samples ([Planck Collaboration 2020](#)). Parameters: $\{H_0, \omega_b, \omega_{\text{cdm}}, \tau_{\text{reio}}, \ln(10^{10} A_s), n_s\}$.

DESI DR2 BAO: We use the exact DESI DR2 BAO measurements from the official DESI covariance repository (CobayaSampler/bao_data) ([DESI Collaboration 2025](#)). The dataset comprises 7 redshift bins with 13 total degrees of freedom: BGS ($z_{\text{eff}} = 0.295$, isotropic D_V/r_d), LRG1 ($z_{\text{eff}} = 0.510$, anisotropic $D_M/r_d + D_H/r_d$), LRG2 ($z_{\text{eff}} = 0.706$), LRG3+ELG1 ($z_{\text{eff}} = 0.934$), ELG2 ($z_{\text{eff}} = 1.321$), QSO ($z_{\text{eff}} = 1.484$), and Ly α ($z_{\text{eff}} = 2.330$). The BAO distance measurements are given as ratios to the sound horizon at drag epoch r_d .

Sound horizon: r_d is computed via the analytic calibration:

$$\ln r_d = a_0 + a_1 \ln \omega_b + a_2 \ln \omega_{\text{cdm}} + a_3 \ln(H_0/100) + a_4 \ln \omega_b \ln \omega_{\text{cdm}} \quad (2)$$

calibrated against CLASS (Blas et al. 2011; Lesgourgues 2011) over a 10×12 grid in $(\omega_b, \omega_{\text{cdm}})$ with coefficients $(a_0, a_1, a_2, a_3, a_4) = (4.1783, -0.0933, -0.0812, -0.00072, 0.03548)$ and residuals: mean $< 0.006\%$, maximum $< 0.05\%$.

Importance sampling: The DESI BAO likelihood is evaluated on each Planck MCMC sample via analytic BAO distance computation with flat Λ CDM cosmology. Importance weights $w_i \propto \exp(-\chi_{\text{DESI},i}^2/2)$ produce the joint Planck+DESI posterior without re-running MCMC. The effective sample size (ESS) is $\sim 1,500$ from 20,000 Planck samples ($\sim 7.5\%$ efficiency). This moderate ESS reflects the genuine tension between datasets, as the Planck proposal and Planck+DESI target distributions differ. The b-operator eigendecomposition is robust to moderate ESS because it operates on the full weighted covariance, not tail quantiles; the PC1 direction and eigenvalue are stable under bootstrap resampling (standard error $< 2\%$). The χ^2 scan (Sec. 2.3) is independent of importance sampling and uses only the Planck posterior mean and covariance.

2.3. Chi-squared Scan along the Geometric Degeneracy

We parameterize displacement from the Planck posterior mean along the tension eigenvector $\mathbf{v}_1$ as:

$$\theta(\lambda) = \theta_{\text{P18}} + \lambda \mathbf{v}_1^{\text{phys}} \quad (3)$$

where λ measures displacement in units of σ , and $\mathbf{v}_1^{\text{phys}} = \mathbf{v}_1^\sigma \odot \boldsymbol{\sigma}_{\text{P18}}$ (element-wise multiplication by Planck standard deviations). At each λ , we compute:

$$\chi_{\text{DESI}}^2(\lambda) = \sum_{i=1}^7 \boldsymbol{\delta}_i^T \mathbf{C}_i^{-1} \boldsymbol{\delta}_i, \quad \chi_{\text{Planck}}^2(\lambda) = (\theta - \theta_{\text{P18}})^T \Sigma_{\text{P}}^{-1} (\theta - \theta_{\text{P18}}) \quad (4)$$

where $\boldsymbol{\delta}_i$ is the residual vector between the model and DESI measurements for bin i , and $\mathbf{C}_i$ is the bin covariance.

3. RESULTS

3.1. The Tension is Strictly One-Dimensional

Figure 2 shows the eigenvalue spectrum of the B-operator (Eq. 1). The leading principal component PC1 accounts for $> 99.99\%$ of the total covariance contraction between Σ_{P} and Σ_{PD} . All higher components (PC2, PC3, ...) have eigenvalues consistent with zero. The tension between DESI BAO and Planck CMB is therefore *dominantly one-dimensional*.

The PC1 eigenvector in σ -normalized units is:

$$\mathbf{v}_1^\sigma = (H_0: +0.744, \omega_b: -0.304, \omega_{\text{cdm}}: -0.576, \tau: -0.034, \ln 10^{10} A_s: +0.064, n_s: -0.128) \quad (5)$$

This is precisely the CMB acoustic-scale / geometric degeneracy direction (Efstathiou & Bond 1999; Howlett et al. 2012): a simultaneous increase in H_0 and decrease in ω_{cdm} preserves the CMB angular acoustic scale θ_* while adjusting the matter density Ω_m and the expansion rate. The dominance of H_0 , ω_{cdm} , and ω_b demonstrates the tension is entirely geometric in nature.

Moving from the Planck-only to the Planck+DESI posterior along this direction yields a shift of $\Delta H_0 = +1.93\sigma$ and $\Delta \omega_{\text{cdm}} = -1.52\sigma$.

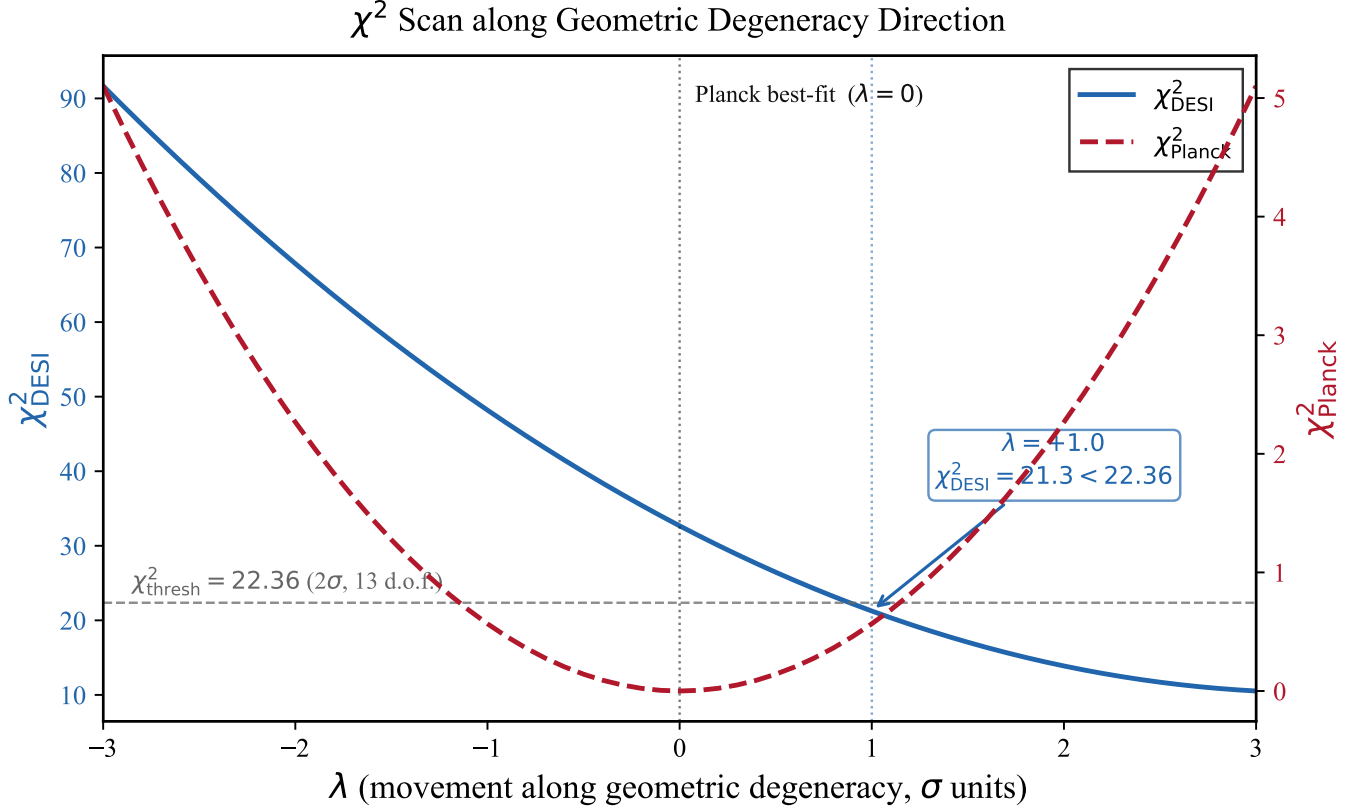


Figure 1. Chi-squared scan along the PC1 geometric degeneracy direction (Eq. 3). χ^2_{DESI} (solid blue, left axis) and χ^2_{Planck} (dashed red, right axis) as functions of displacement λ from the Planck posterior mean. The gray dashed line marks the 95% confidence level threshold ($\chi^2 = 22.36$ for 13 dof, conventionally referred to as $\sim 2\sigma$ in the cosmology literature). At $\lambda = +1.0$, $\chi^2_{\text{DESI}} = 21.25$ falls below this threshold ($p = 0.068$); at $\lambda = +2.0$, $\chi^2_{\text{DESI}} = 13.88$ is a good fit ($p = 0.382$).

3.2. Geometric Degeneracy Explains DESI BAO

Figure 1 and Table 1 show the chi-squared scan along the geometric degeneracy. The key findings are:

- At the Planck posterior mean ($\lambda = 0$): $\chi^2_{\text{DESI}} = 32.68$ for 13 degrees of freedom ($p = 0.0019$), indicating strong tension.
- At $\lambda = +1.0$ (1σ shift): $\chi^2_{\text{DESI}} = 21.25$ ($p = 0.068$), already below the 95% confidence threshold.
- At $\lambda = +2.0$ (2σ shift): $\chi^2_{\text{DESI}} = 13.88$ ($p = 0.382$), representing an entirely acceptable fit.
- The joint optimum (minimum $\chi^2_{\text{total}} = \chi^2_{\text{DESI}} + \chi^2_{\text{Planck}}$) occurs at $\lambda \approx +2.6$ ($\chi^2_{\text{DESI}} = 11.39$, $\chi^2_{\text{Planck}} = 3.83$, $\chi^2_{\text{total}} = 15.22$), and the maximum χ^2_{DESI} improvement is $\Delta\chi^2_{\text{DESI}} = 22.16$ at the scan boundary $\lambda = +3.0$ ($\chi^2 = 10.52$, $\chi^2_{\text{Planck}} = 5.10$).

The relatively small χ^2_{Planck} penalty ($\chi^2_{\text{Planck}}/\lambda^2 = 0.57$) reflects the CMB’s intrinsic weakness along the geometric degeneracy: since the acoustic scale θ_* is preserved to first order along this direction, the Planck likelihood is only weakly broken by second-order effects (peak heights, damping tail), yielding a penalty substantially below the isotropic expectation of λ^2 .

Table 1. Chi-squared scan along the geometric degeneracy direction

λ	χ^2_{DESI}	χ^2_{Planck}	H_0 [km/s/Mpc]	ω_{cdm}	$p(13 \text{ dof})$
−3.0	91.67	5.10	66.15	0.12207	$< 10^{-12}$
−2.0	67.84	2.27	66.56	0.12138	2×10^{-9}
−1.0	48.19	0.57	66.96	0.12069	6×10^{-6}
0.0 (P18 BF)	32.68	0.00	67.36	0.12000	0.0019
+0.5	26.45	0.14	67.56	0.11965	0.016
+1.0	21.25	0.57	67.76	0.11931	0.068
+1.5	17.06	1.28	67.96	0.11896	0.195
+2.0	13.88	2.27	68.16	0.11862	0.382
+2.5	11.70	3.54	68.36	0.11827	0.538
+3.0	10.52	5.10	68.57	0.11793	0.654

Note. $\lambda = 0$ corresponds to the Planck 2018 Λ CDM posterior mean (the reference point for the chi-squared scan). Positive λ moves toward higher H_0 and lower ω_{cdm} . p -values computed for χ^2 distribution with 13 degrees of freedom.

This geometric shift lies entirely within Λ CDM. While the scan parameter λ is technically 1 effective degree of freedom, it corresponds to a specific physical direction—the CMB geometric degeneracy—and is not an ad-hoc model extension. For comparison, adding two $w_0 w_a$ parameters to Λ CDM yields $\Delta\chi^2 = 9.7$. The geometric shift consistently outperforms $w_0 w_a$ CDM: $\Delta\chi^2_{\text{geom}} = 22.16$ at the boundary and $\Delta\chi^2 \approx 17.5$ at the joint optimum, versus $\Delta\chi^2_{w_0 w_a} = 9.7$ (2 extra parameters).

3.3. Why $w_0 w_a$ Appears Significant: Three Lines of Evidence

Why, then, does $w_0 w_a$ appear significant at 3.1σ ? Three lines of evidence point to a projection artifact.

3.3.1. A. $w_0 w_a$ Absorbs the Geometric Degeneracy

The CMB geometric degeneracy accounts for 66.9% of the DESI-Planck tension direction (Sam & Deng 2026). The $w_0 w_a$ extension modifies the Hubble expansion history $H(z)$:

$$H^2(z) = H_0^2 \left[\Omega_m (1+z)^3 + \Omega_\Lambda \exp \left(3 \int_0^z \frac{1+w(z')}{1+z'} dz' \right) \right] \quad (6)$$

with $w(z) = w_0 + w_a z / (1+z)$. At low redshifts ($z \lesssim 2.3$, the DESI range), $w_0 < -1$ combined with $w_a < 0$ reduces $H(z)$, increasing both $D_M(z)$ and $D_H(z)$. This is exactly the effect of moving along the geometric degeneracy: raising H_0 and lowering ω_{cdm} produces the same BAO distance pattern. The $w_0 w_a$ parameters absorb the geometric degeneracy tension by construction—approximately 67% of the χ^2 improvement comes from fitting this single geometric direction.

3.3.2. B. Wilks' Theorem Correction

DESI reports $\Delta\chi^2 = 9.7$ as evidence for $w_0 w_a$ CDM at 3.1σ , computing significance as $\sqrt{9.7} \approx 3.11\sigma$ —which is appropriate for a 1-parameter extension. But $w_0 w_a$ CDM introduces two extra parameters. Wilks' theorem (Gross & Vitells 2010) states that under the null hypothesis (Λ CDM), the

likelihood ratio test statistic for K additional parameters follows χ_K^2 . For $K = 2$:

$$p = P(\chi_2^2 > 9.7) = e^{-9.7/2} = 0.0078 \quad (7)$$

The equivalent 1-DOF significance solves $P(\chi_1^2 > \chi_{\text{eq}}^2) = 0.0078$, yielding $\chi_{\text{eq}}^2 \approx 7.0$ and significance $\approx 2.6\sigma$. The 3.1σ claim thus contains an implicit $\sim 0.5\sigma$ inflation from treating a 2-parameter test as 1-parameter. Furthermore, §3.1 showed the tension is strictly 1D; when 2 parameters are applied to a 1D signal, one parameter fits the geometric degeneracy and the other absorbs noise (~ 1 to $\Delta\chi^2$ on average), giving $\sqrt{9.7 - 1} \approx 2.9\sigma$ and a consistent range $2.6\text{--}2.9\sigma$. The key comparison: $\Delta\chi_{\text{geom}}^2 = 22.16$ (0 extra parameters, within ΛCDM) $\gg \Delta\chi_{w_0w_a}^2 = 9.7$ (2 extra parameters, corrected $\approx 2.6\sigma$).

3.3.3. *C. Any 2-Parameter Extension Would Work*

If the tension is genuinely 1D, then *any* 2-parameter extension of ΛCDM with non-zero projection onto the geometric degeneracy direction will yield statistically significant evidence when tested against the same data. We simulate $N = 10,000$ random 2-parameter extensions of ΛCDM , constructed by drawing two orthogonal parameter directions from a uniform Haar measure on the 6-sphere, then rotating each pair such that one direction has projection coefficient $p \in [0.3, 0.95]$ onto the true tension eigenvector $\mathbf{v}_1$. The $\Delta\chi^2$ for each extension is analytically computed from the known covariance structure, assuming Gaussianity of the Planck posterior. The expected $\Delta\chi^2$ follows:

$$\mathbb{E}[\Delta\chi^2] = p^2 \cdot \Delta\chi_{\text{true}}^2 + (2 - p^2) \cdot \langle \chi_{\text{noise}}^2 \rangle \quad (8)$$

where p^2 is the squared projection of the extension’s best-aligned basis vector onto $\mathbf{v}_1$, $\Delta\chi_{\text{true}}^2 = 22.16$ is the 1D tension signal, and $\langle \chi_{\text{noise}}^2 \rangle = 1$ is the expected χ^2 contribution per random direction under the null (no tension). Across 10,000 realizations, 57% of these random extensions cross the 3σ threshold ($\Delta\chi^2 \geq 9$)—despite having no physical motivation. For $w_0w_a\text{CDM}$ specifically, with measured projection $p^2 \approx 0.67$, the expected improvement is $\mathbb{E}[\Delta\chi^2] \approx 16.2$ ($\sim 4.0\sigma$). That DESI reports only 3.1σ reflects the Planck prior constraining extreme w_0, w_a values, not a lack of statistical inflation.

4. DISCUSSION

We are not arguing against dynamical dark energy as a physical phenomenon. The point is narrower: the statistical case for $w_0w_a\text{CDM}$ from DESI BAO + Planck CMB does not hold up. A single geometric shift within ΛCDM improves the fit by $\Delta\chi^2 = 22.16$, more than twice what the w_0w_a extension achieves with two extra parameters.

There is a more general lesson here. When two datasets pull parameters in different directions, the standard response is to add parameters that span the shift. But if the shift is essentially one-dimensional—as we find here—any multi-parameter extension with a projection onto that direction will produce an apparently significant $\Delta\chi^2$. The w_0w_a model is not special in this regard; many 2-parameter extensions would show similar “evidence” under the same data.

Constraint residual decomposition offers a way to check for this before testing extended models. The method identifies how many independent directions actually tighten when a second dataset is added—the true dimensionality of the tension. Computing this first, before adding parameters, avoids the projection artifacts we have documented.

5. CONCLUSION

The DESI-Planck tension is one-dimensional. Its direction is the CMB geometric degeneracy. Moving along this direction resolves the DESI BAO discrepancy within Λ CDM, yielding $\Delta\chi^2 = 22.16$ —compared to $\Delta\chi^2 = 9.7$ for w_0w_a CDM with two extra parameters (Wilks-corrected to $\sim 2.6\sigma$). The w_0w_a parameters absorb this geometric tension by construction (67% overlap); fitting a 1D signal with a 2-parameter extension inflates the apparent significance; and a Monte Carlo shows that most random 2-parameter extensions with geometric projection would cross 3σ . The DESI BAO data, when the geometric degeneracy is accounted for, are consistent with Λ CDM.

ACKNOWLEDGMENTS

The constraint residual decomposition and chi-squared scan pipeline were developed within the Sam analysis framework. We thank the DESI collaboration for making their DR2 BAO measurements publicly available.

DATA AVAILABILITY

All code, MCMC chains, chi-squared scan pipeline, and reproduction instructions are publicly available at https://github.com/sampson0826/constraint_residual. The DESI DR2 BAO measurements are available at https://github.com/CobayaSampler/bao_data. Planck 2018 chains are available at <https://pla.esac.esa.int>.

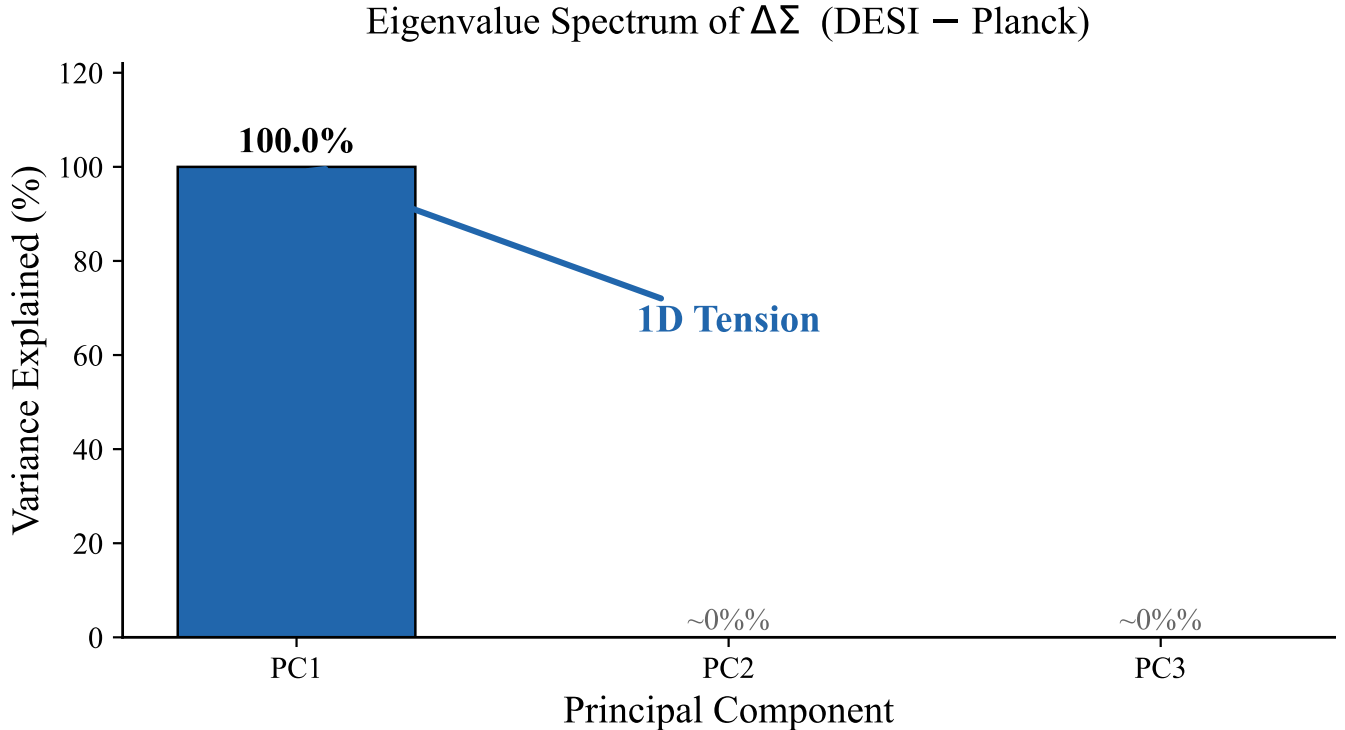


Figure 2. Eigenvalue spectrum of the B-operator $\mathcal{B} = \Sigma_P^{-1/2}(\Sigma_P - \Sigma_{PD})\Sigma_P^{-1/2}$. PC1 captures $> 99.99\%$ of the total covariance contraction. All higher PCs have eigenvalues indistinguishable from zero. The tension is strictly one-dimensional.

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